INCH-POUND
MIL-PRF-8805/31H
21 October 2019
SUPERSEDING
MIL-PRF-8805/31G
14 March 2003

PERFORMANCE SPECIFICATION SHEET

SWITCH, SENSITIVE, DPDT, PLUNGER, 10 AMPERES, UNSEALED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-PRF-8805.

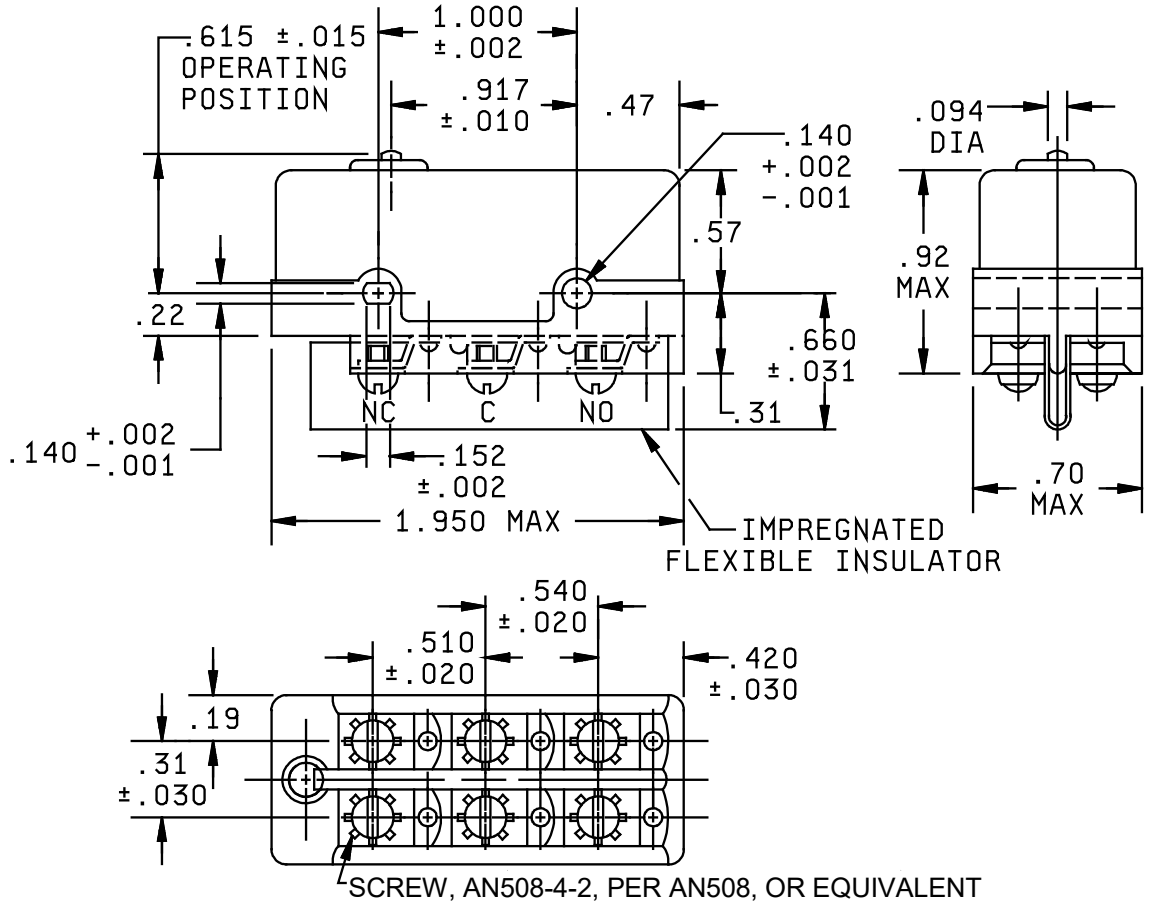


FIGURE 1. Dimensions and configurations.

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Inches	mm	Inches	mm	Inches	mm
.001	0.03	.152	3.86	.57	14.5
.002	0.05	.19	4.8	.615	15.62
.010	0.25	.22	5.6	.660	16.76
.020	0.51	.31	7.9	.70	17.8
.030	0.76	.420	10.67	.917	23.29
.031	0.79	.47	11.9	.92	23.4
.094	2.39	.510	12.95	1.000	25.40
.139	3.53	.540	13.72	1.950	49.43
.140	3.56				

NOTES:
1. Dimensions are in inches.
2. Unless otherwise stated, tolerance is ±.015 (0.38 mm).
3. Metric equivalents are given for general information only.
4. Exact shape of switch optional, provided dimensions specified are not exceeded.
5. Mounting holes shall accept .139 maximum diameter pins or screws on 1,000 ±.002 centers.
6. Alternative base metals and protective finishes, as approved by the qualifying activity, may be utilized for terminal screw and lock washer material. Dimensions shall be in accordance with the referenced hardware specifications.

FIGURE 1. Dimensions and configurations (continued).

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REQUIREMENTS:

Dimensions and configurations: See figure 1.

Enclosure design: 1 (unsealed).

Temperature characteristic: 1 (-55°C to +85°C).

Shock type: (30 g's) in accordance with MIL-STD-202-213, test condition J.

Vibration grade: 1 (10 to 55 Hz) in accordance with MIL-STD-202-201.

Weight: 1.5 ounces maximum.

Operating characteristics:

Actuating force: 20 ounces maximum.

Release force: 2 ounces minimum.

Movement differential: .040 inch to .060 inch (1.02 mm to 1.52 mm).

Overtravel: .005 inch (0.13 mm) minimum.

Pretravel: .075 inch (1.91 mm) maximum.

Contact resistance: Not applicable.

Dielectric withstanding voltage:

Sea level: 1,000 V rms.
Altitude: 400 V rms at 50,000 feet.

Mechanical endurance: 50,000 cycles.

Electrical endurance: 25,000 cycles.

Temperature rise: 65°C.

Electrical ratings: See table I.

Part or Identifying Number (PIN): MS25008-1.

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TABLE I. Electrical ratings.

Load	Sea level 28 V dc	50,000 feet 28 V dc
	amperes	amperes
Resistive	10	10
Inductive	5	2
Motor	5	5

Reference documents:
MIL-PRF-8805
MIL-STD-202-201
MIL-STD-202-213
AN508
NASM35333

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5930-2019-052)

Review activities:
Army - AR, AV, MI
Navy - AS, OS, MC
Air Force - 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/ .

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